



Submitted via email

Office of Science and Technology Policy
Executive Office of the President
Eisenhower Executive Office Building
1650 Pennsylvania Avenue
Washington, D.C. 20504

March 15, 2025

Re: Notice of Request for Information (RFI) on Development of an Artificial Intelligence (AI) Action Plan

The Digital Energy Council (DEC) appreciates the opportunity to respond to this Request for Information (RFI) regarding the Office of Science and Technology Policy (OSTP) Development of an Artificial Intelligence Action Plan. With respect to the initiative, we discuss below additional considerations for addressing energy challenges, particularly as AI applications proliferate and energy demand increases.

About the Digital Energy Council

The Digital Energy Council (DEC) is a non-profit advocacy organization whose preeminent members work at the forefront of the energy and technology industries. DEC was founded to shape the future of energy use and inform policymakers about the important cross-section between the energy industry and the digital applications driving a new economy. As society becomes increasingly digital, the energy sector must evolve to keep pace. It is essential for the energy ecosystem to embrace new technologies and adapt to meet growing demand.

Need for Action

The U.S. energy system is experiencing significant demand driven by the 21st century digital economy. The energy sector has an interest in exploring innovative methods of leveraging energy resources to meet demand associated with digital technologies. Data centers currently consume approximately 2-3% of the total electricity in the United States, with energy demand projected to rise substantially in conjunction with buildout of large-scale computing facilities.¹

In the context of artificial intelligence (AI), bitcoin mining, and high-performance computing (HPC), the concept of “digital energy” represents the synergy between the broader energy sector and the technologies driving the digital economy that require significant computational power. These new technologies can work with our energy systems efficiency, reliability, and can even foster resource development in certain situations. To meet the rising demand for energy, it is imperative that we consider innovative solutions. Leveraging AI, HPC, and bitcoin mining presents the Trump Administration with an opportunity to use all tools and technologies at our

¹ <https://www.eia.gov/todayinenergy/detail.php?id=62409>

disposal to advance the United States' digital energy agenda with any and all fuel sources. Energy dominance will be the key to asserting the United States' technology dominance.

Overcoming Bottlenecks in the Energy Transmission System

A critical near-term action for the administration's whole-of-government AI strategy should be interconnection queue reform to address the rapidly growing energy demands of AI and HPC. As the administration seeks to foster a business environment where new AI-driven projects can break ground quickly, it must tackle the persistent bottlenecks in the U.S. transmission system. Currently, interconnection queues are lengthy, opaque, and overburdened, delaying critical energy infrastructure necessary to support AI workloads. Without streamlined reforms that enhance transparency, efficiency, and coordination among grid operators and developers, the United States risks falling behind in the global race for digital energy leadership. AI innovation depends on scalable, reliable, and sustainable power, but funded projects are stuck waiting for years due to outdated interconnection processes.

The Growing Concern of Foreign Control over Digital Infrastructure

The United States' adversaries are increasingly offering financial incentives, strategic advantages, and favorable regulatory treatment to attract digital infrastructure companies to relocate their operations overseas. These tactics pose a significant concern for U.S. national security, economic stability, privacy, and cyber resilience, particularly for the energy industry, which depends on secure data systems for operations.

The United States' adversaries use tax breaks, grants, and subsidies to lure technology infrastructure construction. This strategy reduces U.S. influence over critical infrastructure and has raised concerns regarding security and national interests. Foreign adversaries can gain physical access to sensitive data through data centers, which could allow them to influence operations, monitor communications, or manipulate data.

Many countries with weak or lax data protection laws present risks for the confidentiality of U.S. data stored in their data centers. Countries like Russia, China, India, and the UAE have been identified as potential vulnerabilities. Control over data centers also provides adversaries with strategic advantages, such as a platform for cyber-espionage, the ability to disrupt U.S. services, and the potential to target critical sectors like energy.

Use AI and HPC to Support Energy Infrastructure

AI and HPC technologies have revolutionized various industries, enabling us to process vast amounts of data efficiently and solve complex problems. These technologies can be utilized to optimize energy production, distribution, and consumption. By employing AI algorithms and HPC systems, we can enhance predictive maintenance in energy infrastructure, improve grid management, and optimize energy usage in buildings and industrial processes. Furthermore, AI-driven simulations can facilitate the development of advanced energy technologies.

Leverage Bitcoin Mining to Improve Energy Grid Efficiency and Resilience

By integrating Bitcoin mining operations with energy grids, excess energy generated from intermittent sources can be utilized for mining activities during off-peak hours, thus maximizing resource utilization and minimizing wastage. Mining operations have also demonstrated that their unique “flexible load” can be leveraged during emergency situations by curtailing demand.

Recommendations

A whole-of-government approach to energy and technology is needed. Accordingly, the Trump administration should:

- Amplify expertise in critical and emerging technology and leverage these capabilities to solve critical science, energy, and security challenges facing the Nation;
- Coordinate equities in critical and emerging technologies across diverse program portfolios, including the DOE National Labs, to support and inform policy making;
- Partner with industry (energy & tech) and academia to advance breakthroughs and accelerate integration of energy technologies into the U.S. commercial sector.
- Revisit the Trump Administration's American AI Initiative via Executive Order 13859 and explore incorporating an added emphasis on energy as a top priority. Develop a regulatory guidance to govern the deployment and operation of AI, HPC, and Bitcoin mining facilities within the energy sector. A pro-business and deregulatory approach encouraged innovation in AI and data centers.
- Allocate resources towards research and development initiatives aimed at exploring the potential of AI, HPC, and Bitcoin mining as energy resources to advance industries of the future. Foster collaborations with academic institutions, industry partners, and technology innovators to accelerate progress in this area.
- Initiate pilot projects to test the feasibility and effectiveness of integrating AI, HPC, and Bitcoin mining with existing energy infrastructure. Incentivize infrastructure development in coordination with our energy systems.
- Educate stakeholders and the general public about the potential benefits of digital energy, leveraging AI, HPC, and Bitcoin mining as resources. Address concerns related to energy consumption, sustainability, and economic viability through transparent communication and outreach efforts.

Conclusion

The Digital Energy Council firmly believes that technological advancements can only progress as far as the energy foundation upon which they are built. The United States cannot maintain its leadership in innovation without a robust, reliable, and forward-thinking energy strategy. Ensuring a strong energy foundation is not just an economic necessity—it is a prerequisite for continued innovation and global leadership.

We appreciate the opportunity to respond to this RFI and look forward to serving as a resource for these important and complex issue areas. Please do not hesitate to reach out at info@digitalenergycouncil.org if you have any questions.

